Muneer Al-Khasawneh & Muhanad Al-Khasawneh - GROUP 24

CS 5165 – Introduction To Cloud Computing
Final Group Project: Data Science and Analytics Using Azure Cloud Computing Technologies

Final Project Group 24 Results

Machine Learning Models Overview

Linear Regression is a simple algorithm used to predict continuous values by modeling the relationship between variables. It can be used to estimate customer spending trends over time.

Random Forest is an ensemble learning method that builds multiple decision trees and combines their outputs to improve prediction accuracy. In this project, it was used to classify whether a customer is a high spender based on purchasing behavior.

Gradient Boosting is another ensemble method that builds models sequentially, improving errors from previous models. It is useful for making highly accurate predictions in complex datasets.

For Customer Lifetime Value (CLV), predictive models can estimate long-term customer spending by analyzing historical purchase behavior, helping businesses identify high-value customers.

Application Design and Features

This project was deployed using Azure Web App Services and provides an interactive dashboard for analyzing retail data.

The application includes:

- A login page that requires a username, password, and email
- A dashboard displaying KPIs such as total sales, average spend, and total units
- Interactive charts showing trends over time, department sales, and regional performance
- A search feature allowing users to filter data by household ID
- A sample data pull that integrates Transactions, Households, and Products datasets
- Sorting of results based on required fields

Data Storage and Loading

The dataset is stored using Azure-hosted storage and loaded dynamically into the application using CSV-based ingestion. This approach allows efficient processing of large datasets while maintaining low cost.

Machine Learning Application (Basket Analysis)

Basket analysis was implemented by identifying frequently purchased product pairs within the same transaction. This helps identify cross-selling opportunities by showing which products are commonly bought together.

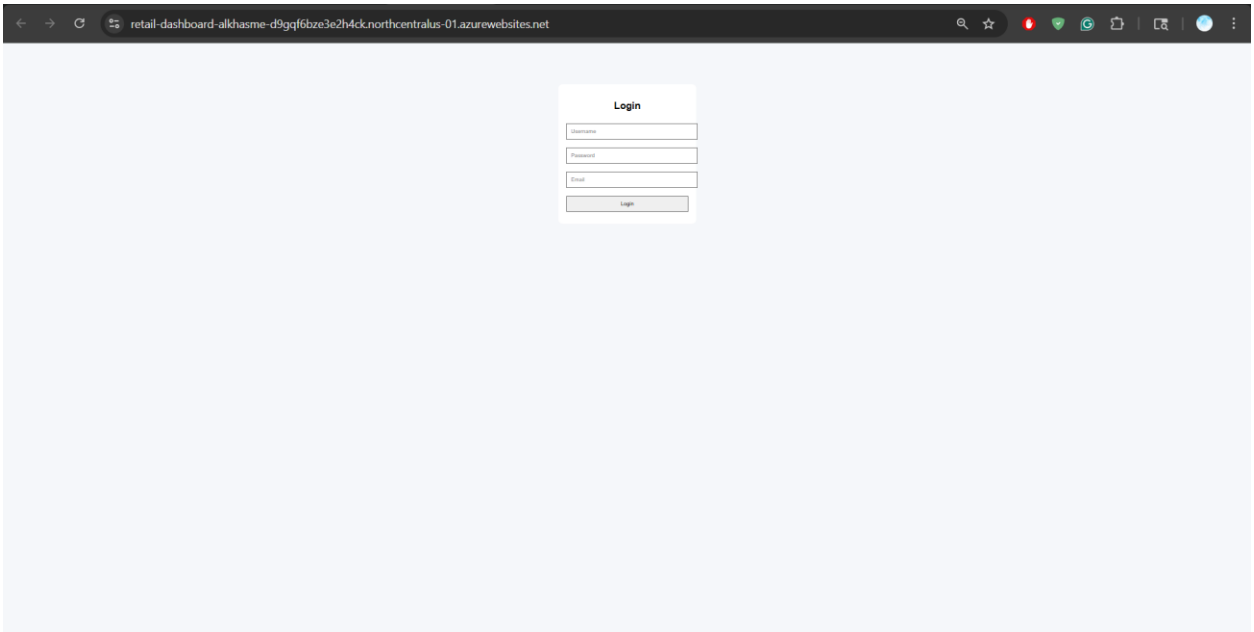
A Random Forest model was also used to classify customers as high or low spenders based on purchase behavior.

Churn Prediction

Churn risk was estimated by identifying customers whose total spending falls below the average. These customers are considered at risk of disengaging and can be targeted for retention strategies.

Conclusion

This project demonstrates how cloud computing, data engineering, and machine learning can be integrated to generate actionable insights in a retail setting. The dashboard provides an easy-to-use interface for exploring customer behavior and business trends.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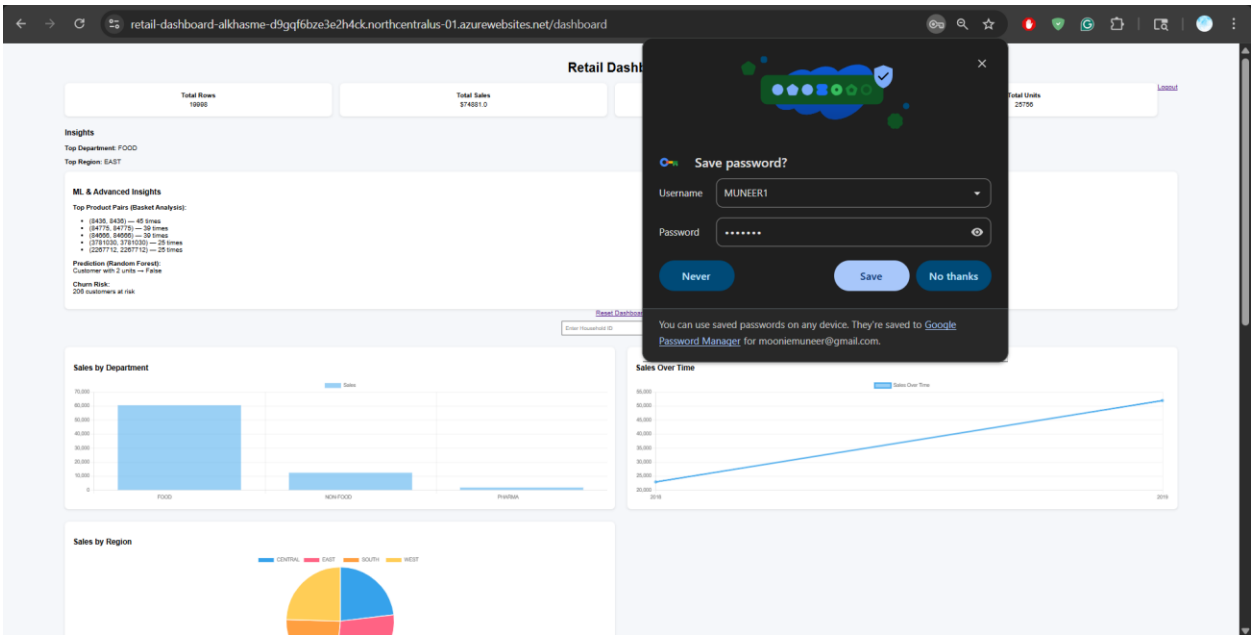
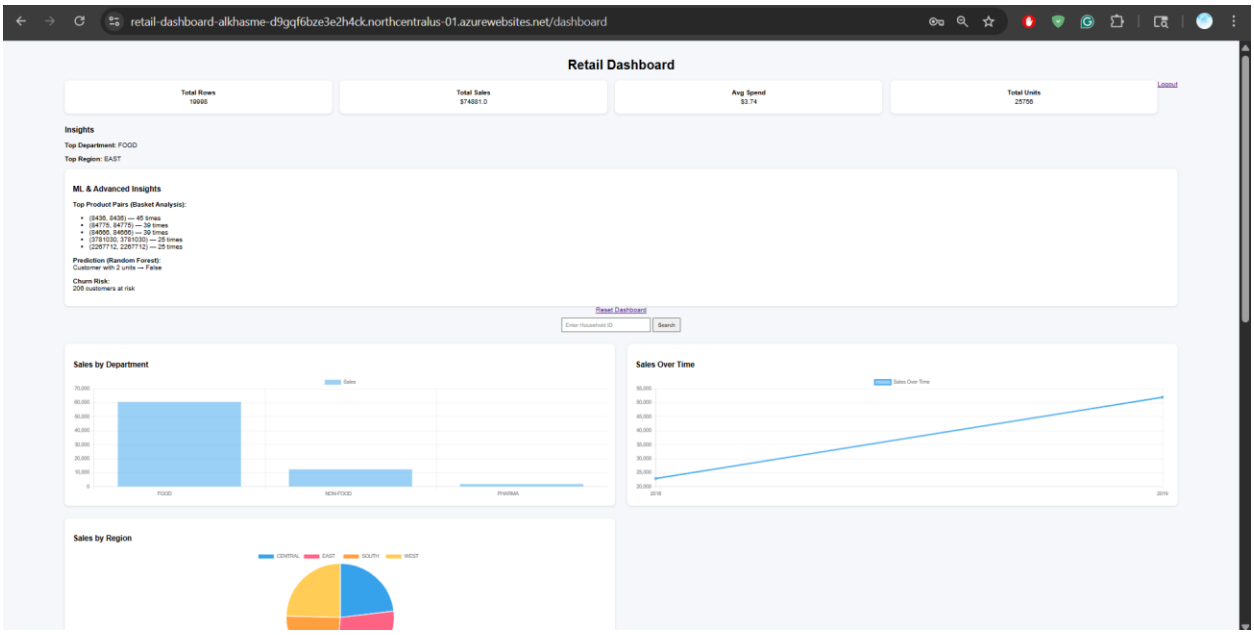
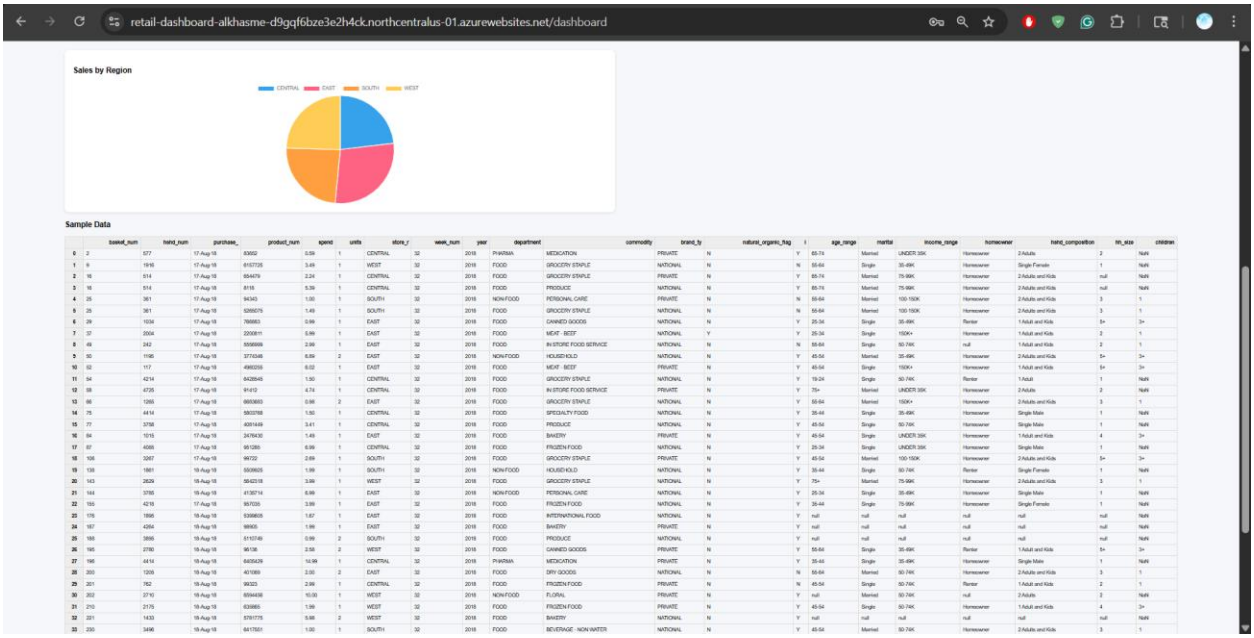
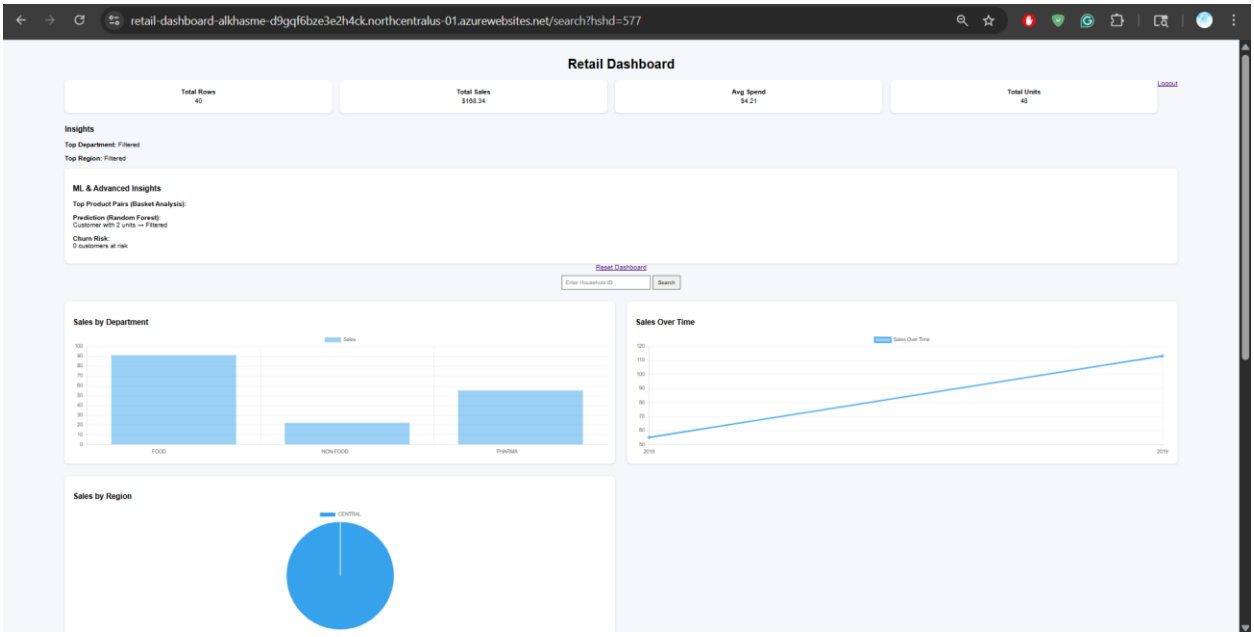
Login
Username
Password
Email
Login

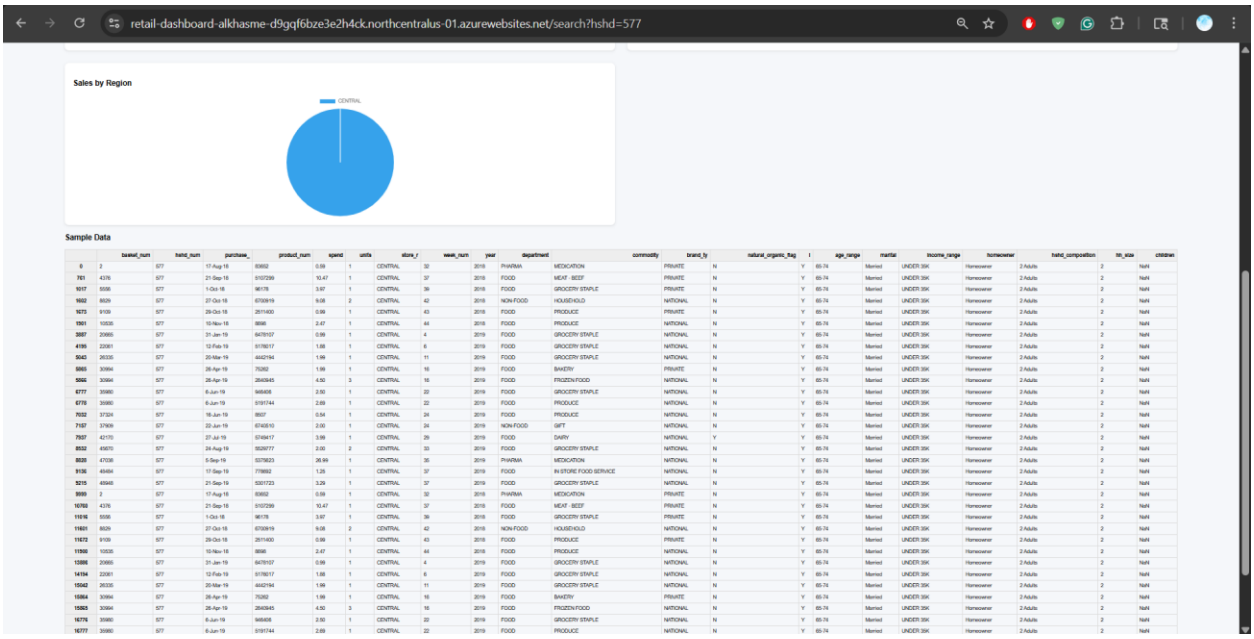

retail-dashboard-alkhasme-d9gqf6bze3e2h4ck.northcentralus-01.azurewebsites.net/dashboard
Retail Dashb
Total Rows
19998
Total Sales
574881.0
Total Units
25755
Logout
Insights
Top Department: FOOD
Top Region: EAST
ML & Advanced Insights
Top Product Pairs (Basket Analysis):
(8435, 8436) — 45 times
(84775, 84775) — 39 times
(84866, 84866) — 39 times
(3781030, 3781030) — 25 times
(2267712, 2267712) — 25 times
Prediction (Random Forest):
Customer with 2 units — False
Churn Risk:
206 customers at risk
Reset Dashboard
Enter Household ID
Save password?
Username
MUNEER1
Password
Never
Save
No thanks
You can use saved passwords on any device. They're saved to Google Password Manager for mooniemuneer@gmail.com.
Sales by Department
Sales
FOOD
NON-FOOD
PHARMA
Sales Over Time
Sales Over Time
2018
2019
Sales by Region
CENTRAL
EAST
SOUTH
WEST





	basket_num	hshd_num	purchase_	product_num	spend	units	store_r	week_num	year	department	commodity	brand_ty	natural_organic_flag	l	age_range	marital	income_range	homeowner	hshd_composition	hh_size	children
0	2	577	17-Aug-18	83602	0.59	1	CENTRAL	32	2018	PHARMA	MEDICATION	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
1	9	1916	17-Aug-18	6107735	3.49	1	WEST	32	2018	FOOD	GROCERY STAPLE	NATIONAL	N	N	55-64	Single	35-49K	Homeowner	Single Female	1	NaN
2	16	514	17-Aug-18	664079	2.24	1	CENTRAL	32	2018	FOOD	GROCERY STAPLE	PRIVATE	N	Y	65-74	Married	75-99K	Homeowner	2 Adults and Kids	null	NaN
3	16	514	17-Aug-18	8115	5.39	1	CENTRAL	32	2018	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	75-99K	Homeowner	2 Adults and Kids	null	NaN
4	25	361	17-Aug-18	94343	1.00	1	SOUTH	32	2018	NON-FOOD	PERSONAL CARE	PRIVATE	N	N	55-64	Married	100-150K	Homeowner	2 Adults and Kids	3	1
5	25	361	17-Aug-18	5295075	1.49	1	SOUTH	32	2018	FOOD	GROCERY STAPLE	NATIONAL	N	N	55-64	Married	100-150K	Homeowner	2 Adults and Kids	3	1
6	29	1034	17-Aug-18	766653	0.99	1	EAST	32	2018	FOOD	CANNED GOODS	NATIONAL	N	Y	25-34	Single	35-49K	Renter	1 Adult and Kids	5+	3+
7	37	2004	17-Aug-18	2000811	0.99	1	EAST	32	2018	FOOD	MEAT - BEEF	NATIONAL	Y	Y	25-34	Single	150K+	Homeowner	1 Adult and Kids	2	1
8	49	242	17-Aug-18	5556999	2.99	1	EAST	32	2018	FOOD	IN-STORE FOOD SERVICE	NATIONAL	N	N	55-64	Single	50-74K	null	1 Adult and Kids	2	1
9	50	1196	17-Aug-18	3774348	6.69	2	EAST	32	2018	NON-FOOD	HOUSEHOLD	NATIONAL	N	Y	45-54	Married	35-49K	Homeowner	2 Adults and Kids	5+	3+
10	52	117	17-Aug-18	4960255	6.02	1	EAST	32	2018	FOOD	MEAT - BEEF	PRIVATE	N	Y	45-54	Single	150K+	Homeowner	1 Adult and Kids	5+	3+
11	54	4214	17-Aug-18	6428545	1.50	1	CENTRAL	32	2018	FOOD	GROCERY STAPLE	NATIONAL	N	Y	19-24	Single	50-74K	Renter	1 Adult	1	NaN
12	58	4735	17-Aug-18	91412	4.74	1	CENTRAL	32	2018	FOOD	IN-STORE FOOD SERVICE	PRIVATE	N	Y	75+	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
13	66	1265	17-Aug-18	6803693	0.99	2	EAST	32	2018	FOOD	GROCERY STAPLE	NATIONAL	N	Y	55-64	Married	150K+	Homeowner	2 Adults and Kids	3	1
14	75	4414	17-Aug-18	5803768	1.50	1	CENTRAL	32	2018	FOOD	SPECIALTY FOOD	NATIONAL	N	Y	35-44	Single	35-49K	Homeowner	Single Male	1	NaN
15	77	3758	17-Aug-18	4061449	3.41	1	CENTRAL	32	2018	FOOD	PRODUCE	NATIONAL	N	Y	45-54	Single	50-74K	Homeowner	Single Male	1	NaN
16	84	1015	17-Aug-18	3479430	1.49	1	EAST	32	2018	FOOD	BAKERY	PRIVATE	N	Y	45-54	Single	UNDER 35K	Homeowner	1 Adult and Kids	4	3+
17	87	4069	17-Aug-18	951285	6.99	1	CENTRAL	32	2018	FOOD	FROZEN FOOD	NATIONAL	N	Y	25-34	Single	UNDER 35K	Homeowner	Single Male	1	NaN
18	106	3267	17-Aug-18	96722	2.69	1	SOUTH	32	2018	FOOD	GROCERY STAPLE	PRIVATE	N	Y	45-54	Married	100-150K	Homeowner	2 Adults and Kids	5+	3+
19	138	1861	18-Aug-18	5509925	1.99	1	SOUTH	32	2018	NON-FOOD	HOUSEHOLD	NATIONAL	N	Y	35-44	Single	50-74K	Renter	Single Female	1	NaN
20	143	2829	18-Aug-18	5842518	3.99	1	WEST	32	2018	FOOD	GROCERY STAPLE	NATIONAL	N	Y	75+	Married	75-99K	Homeowner	2 Adults and Kids	3	1
21	144	3785	18-Aug-18	4130714	6.99	1	EAST	32	2018	NON-FOOD	PERSONAL CARE	NATIONAL	N	Y	25-34	Single	35-49K	Homeowner	Single Male	1	NaN
22	155	4219	17-Aug-18	957035	3.99	1	EAST	32	2018	FOOD	FROZEN FOOD	NATIONAL	N	Y	35-44	Single	75-99K	Homeowner	Single Female	1	NaN
23	176	1896	18-Aug-18	5398605	1.67	1	EAST	32	2018	FOOD	INTERNATIONAL FOOD	NATIONAL	N	Y	null	null	null	null	null	null	NaN
24	187	4264	18-Aug-18	98905	1.99	1	EAST	32	2018	FOOD	BAKERY	PRIVATE	N	Y	null	null	null	null	null	null	NaN
25	188	3886	18-Aug-18	5115749	0.99	2	SOUTH	32	2018	FOOD	PRODUCE	NATIONAL	N	Y	null	null	null	null	null	null	NaN
26	195	2780	18-Aug-18	96156	2.59	2	WEST	32	2018	FOOD	CANNED GOODS	PRIVATE	N	Y	55-64	Single	35-49K	Renter	1 Adult and Kids	5+	3+
27	196	4414	18-Aug-18	6405429	16.99	1	CENTRAL	32	2018	PHARMA	MEDICATION	PRIVATE	Y	Y	35-44	Single	35-49K	Homeowner	Single Male	1	NaN
28	200	1209	18-Aug-18	4011669	2.00	2	EAST	32	2018	FOOD	DRY GOODS	NATIONAL	N	N	55-64	Married	50-74K	Homeowner	2 Adults and Kids	3	1
29	201	762	18-Aug-18	99323	2.99	1	CENTRAL	32	2018	FOOD	FROZEN FOOD	PRIVATE	N	N	45-54	Single	50-74K	Renter	1 Adult and Kids	2	1
30	202	2710	18-Aug-18	6594458	10.00	1	WEST	32	2018	NON-FOOD	FLORAL	PRIVATE	N	Y	null	Married	50-74K	null	2 Adults	2	NaN
31	210	2175	18-Aug-18	639865	1.99	1	WEST	32	2018	FOOD	FROZEN FOOD	PRIVATE	N	Y	45-54	Single	50-74K	Homeowner	1 Adult and Kids	4	3+
32	221	1433	18-Aug-18	5761775	5.68	2	WEST	32	2018	FOOD	BAKERY	NATIONAL	N	Y	null	null	null	null	null	null	NaN
33	230	3496	18-Aug-18	6417551	1.00	1	SOUTH	32	2018	FOOD	BEVERAGE - NON WATER	NATIONAL	N	Y	45-54	Married	50-74K	Homeowner	2 Adults and Kids	3	1





	basket_num	hshd_num	purchase_	product_num	spend	units	store_r	week_num	year	department	commodity	brand_ty	natural_organic_flag	l	age_range	marital	income_range	homeowner	hshd_composition	hh_size	children
0	2	577	17-Aug-18	63652	0.59	1	CENTRAL	32	2018	PHARMA	MEDICATION	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
701	4376	577	21-Sep-18	5107299	10.47	1	CENTRAL	37	2018	FOOD	MEAT - BEEF	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
1017	5556	577	1-Oct-18	96175	3.97	1	CENTRAL	39	2018	FOOD	GROCERY STAPLE	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
1602	8529	577	27-Oct-18	6709919	9.08	2	CENTRAL	42	2018	NON-FOOD	HOUSEHOLD	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
1673	9109	577	29-Oct-18	2511400	0.99	1	CENTRAL	43	2018	FOOD	PRODUCE	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
1501	10535	577	10-Nov-18	8898	2.47	1	CENTRAL	44	2018	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
3557	20665	577	31-Jan-19	6478107	0.99	1	CENTRAL	4	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
4199	22061	577	12-Feb-19	5178217	1.68	1	CENTRAL	6	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
5043	26335	577	20-Mar-19	4442194	1.99	1	CENTRAL	11	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
5865	30994	577	26-Apr-19	75262	1.99	1	CENTRAL	16	2019	FOOD	BAKERY	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
5866	30994	577	26-Apr-19	2840945	4.50	3	CENTRAL	16	2019	FOOD	FROZEN FOOD	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
6777	35980	577	6-Jun-19	948468	2.50	1	CENTRAL	22	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
6778	35980	577	6-Jun-19	5191744	2.69	1	CENTRAL	22	2019	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
7032	37324	577	16-Jun-19	8897	0.54	1	CENTRAL	24	2019	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
7157	37909	577	22-Jun-19	6740510	2.00	1	CENTRAL	24	2019	NON-FOOD	GIFT	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
7937	42170	577	27-Jul-19	5749417	3.99	1	CENTRAL	29	2019	FOOD	DAIRY	NATIONAL	Y	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
8532	45670	577	24-Aug-19	5526777	2.00	2	CENTRAL	33	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
8820	47036	577	5-Sep-19	5375823	26.99	1	CENTRAL	35	2019	PHARMA	MEDICATION	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
9156	49484	577	17-Sep-19	778862	1.25	1	CENTRAL	37	2019	FOOD	IN STORE FOOD SERVICE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
9215	49948	577	21-Sep-19	5301723	3.29	1	CENTRAL	37	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
9999	2	577	17-Aug-18	63652	0.59	1	CENTRAL	32	2018	PHARMA	MEDICATION	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
10703	4376	577	21-Sep-18	5107299	10.47	1	CENTRAL	37	2018	FOOD	MEAT - BEEF	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
11016	5556	577	1-Oct-18	96175	3.97	1	CENTRAL	39	2018	FOOD	GROCERY STAPLE	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
11601	8529	577	27-Oct-18	6709919	9.08	2	CENTRAL	42	2018	NON-FOOD	HOUSEHOLD	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
11672	9109	577	29-Oct-18	2511400	0.99	1	CENTRAL	43	2018	FOOD	PRODUCE	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
11500	10535	577	10-Nov-18	8898	2.47	1	CENTRAL	44	2018	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
13886	20665	577	31-Jan-19	6478107	0.99	1	CENTRAL	4	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
14194	22061	577	12-Feb-19	5178217	1.68	1	CENTRAL	6	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
15042	26335	577	20-Mar-19	4442194	1.99	1	CENTRAL	11	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
15864	30994	577	26-Apr-19	75262	1.99	1	CENTRAL	16	2019	FOOD	BAKERY	PRIVATE	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
15863	30994	577	26-Apr-19	2840945	4.50	3	CENTRAL	16	2019	FOOD	FROZEN FOOD	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
16776	35980	577	6-Jun-19	948468	2.50	1	CENTRAL	22	2019	FOOD	GROCERY STAPLE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN
16777	35980	577	6-Jun-19	5191744	2.69	1	CENTRAL	22	2019	FOOD	PRODUCE	NATIONAL	N	Y	65-74	Married	UNDER 35K	Homeowner	2 Adults	2	NaN